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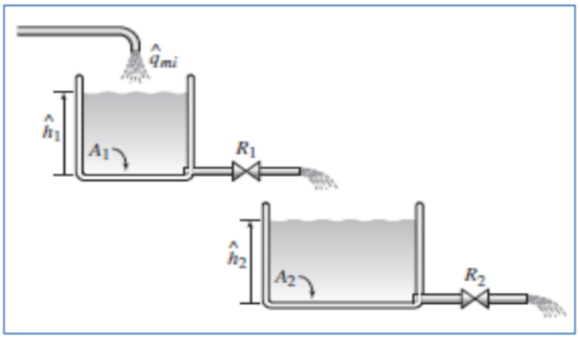
Question (1): [10 Marks]

a) For the following liquid system:
   i. Derive the governing differential equations. **(2 marks)**
   ii. Find the transfer function between q_i and h_2 of the two tanks. **(2 marks)**
   iii. Represent the transfer functions of the two tanks in a block diagram. **(1 mark)**



i) $\hat{q}_{mi} - \frac{h_1}{R_1} = A_1 \dot{h}_1 \xrightarrow[\times R_1]{\mathcal{L}} R_1 Q_{mi}(s) - H_1(s) = sAR_1 H_1(s)$

$\frac{h_1}{R_1} - \frac{h_2}{R_2} = A_2 \dot{h}_2 \xrightarrow[\times R_1 R_2]{\mathcal{L}} R_2 H_1(s) - R_1 H_2(s) = A_2 R_1 R_2 s H_2(s)$

ii) $R_1 Q_{mi}(s) = (sAR_1 + 1) H_1(s) \rightarrow \frac{H_1(s)}{Q_{mi}(s)} = \frac{R_1}{sAR_1 + 1}$

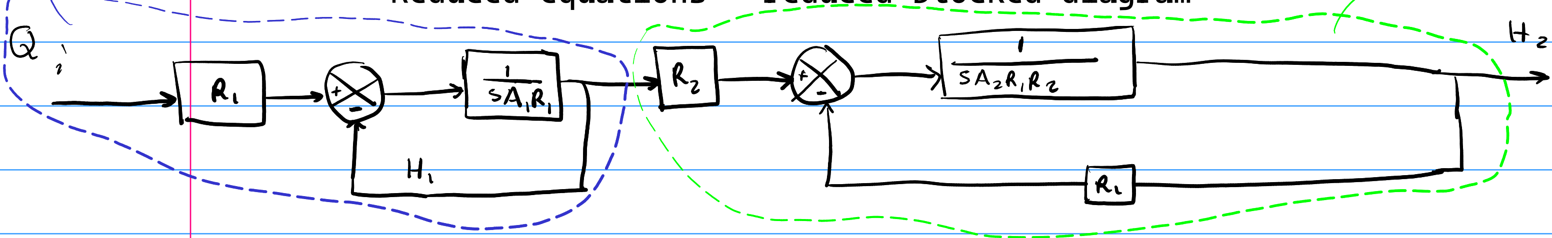
$R_2 H_1(s) = (sAR_1R_2 + R_1) H_2(s) \rightarrow \frac{H_2(s)}{H_1(s)} = \frac{R_2}{sAR_1R_2 + R_1}$

$$\frac{H_2(s)}{Q_{mi}(s)} = \frac{H_1(s)}{Q_{mi}(s)} \times \frac{H_2(s)}{H_1(s)} = \frac{R_1 R_2}{(sAR_1+1)(sAR_1R_2+R_1)}$$

TANK 1

Tank 2

Reduced equations = reduced blocked diagram



iii) $\frac{H_1(s)}{Q_{mi}(s)} = R_1 \cdot \frac{1}{sA_1R_1 + 1} \rightarrow \frac{H_1(s)}{Z(s)} \cdot \frac{Z(s)}{Q_{mi}(s)}$

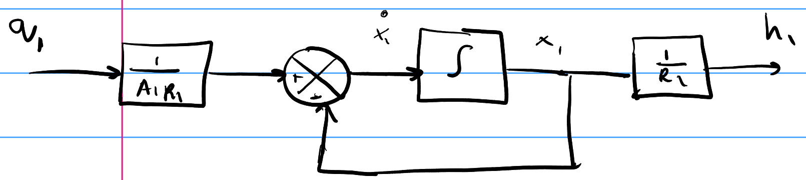
$\frac{H_1(s)}{Z(s)} = R_1 \rightarrow h_1(t) = R_1 z$ $\qquad$ $\frac{Z(s)}{Q_{mi}(s)} = \frac{1}{sA_1R_1 + 1} \rightarrow q = A_1R_1\dot{z} + z$

$x_1 = z \rightarrow \dot{x}_1 = \dot{z}$

$h_1 = R_1 x_1$ $\qquad$ $\dot{x}_1 = \frac{1}{A_1R_1} q - \frac{1}{A_1R_1} x_1$

$x_1 = z \rightarrow \dot{x}_1 = \dot{z}$

$h_1 = R_1 x_1$ $\qquad \dot{x}_1 = \frac{1}{A_1 R_1} q - \frac{1}{A_1 R_1} x_1$



same for $\frac{H_2(s)}{H_1(s)}$